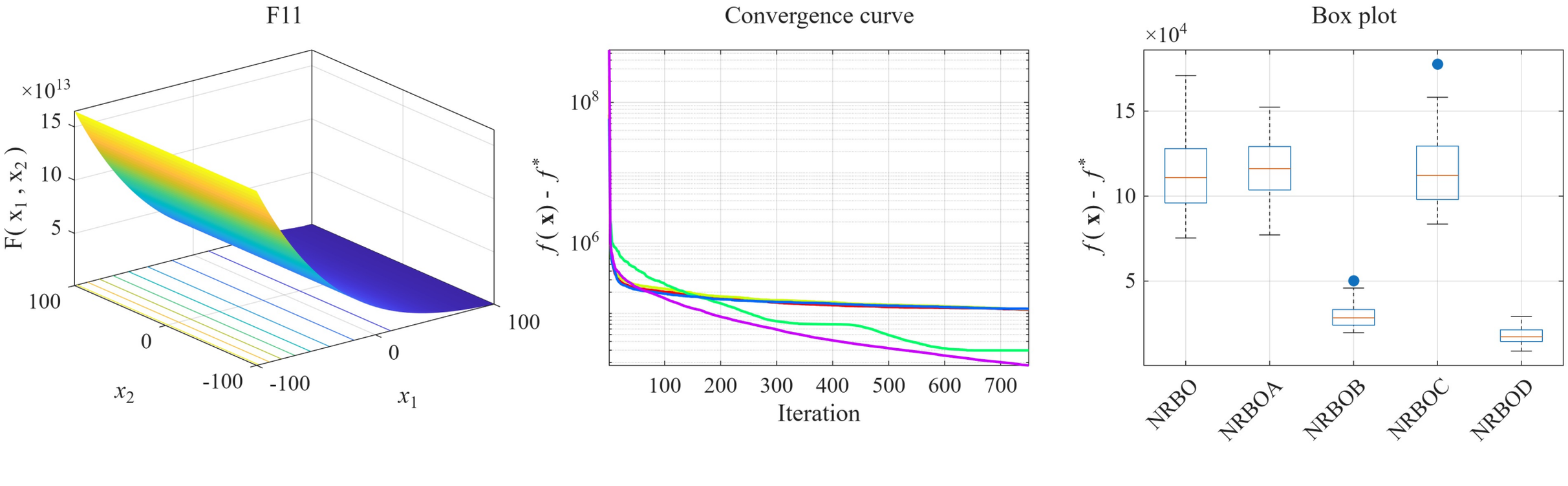

F11
×10^13
15
10
5
F( x1 , x2 )
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^8
10^6
f( x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
15
10
5
f( x) - f*
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD